UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

Abstract

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The measurement of Higgs production cross section in the four-lepton final state at CMS

by Tahir JAVAID

The measurement of properties of Higgs boson is presented in its four-lepton decay channel. The fiducial cross sections (integrated and differential) of the Higgs boson production via $H \to 4\ell$ decays ($\ell = \mathrm{e}$, μ) are measured using the data collected with CMS detector in pp collisions at $\sqrt{s} = 13$ TeV . The measurements use Higgs mass $m_H = 125.09$ GeV and correspond to an integrated luminosity of 137.1 fb^{-1}at 13 TeV(Full Run 2 data). These measurements provide a direct test of perturbative QCD calculations in the Higgs sector of Standard Model at the TeV energy scale.

In order to make all the measurements independent of how Higgs is produced, all the cross sections are extracted within a fiducial phase space defined by the selections on lepton kinematics and event topology. The integrated fiducial cross sections are measured to be $2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\mathrm{stat.})^{+0.24}_{-0.19}(\mathrm{syst.})\ \mathrm{fb}$ at 13 TeV. The differential fiducial cross sections are presented as a function of several kinematic observables which are sensitive to the Higgs-boson production mechanisms: rapidity and transverse momentum of the four-lepton system, associated jet multiplicity and transverse momentum of the leading jet. These measurements are important to test the Standard Model predictions in the Higgs sector and can be used to constrain phase spaces of many theories beyond the Standard Model.

All the integrated and differential measurements are compared with the theoretical predictions based on the Standard Model and found to be consistent within their uncertainties.